Cosmology: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

30 worked solutions · independent explanations · source attribution

Read a solution after making a genuine attempt. The difficult original challenges include an **Idea** paragraph explaining how to choose the method, an explicit **IOAA connection**, and final consistency checks. These are independent MAIS PAC solutions, not official marking schemes. Rounded benchmark parameters are teaching choices, not claims about the latest observational consensus.

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Foundations: worked solutions

Solution 1 — A rate disguised as a velocity. Return to problem →

Idea. A kilometre per second per megaparsec is a fractional rate. Convert the megaparsec to kilometres; the kilometres then cancel.

(a)

$$H_0 = \frac{70}{3.0856776 \times 10^{19}}\ \mathrm{s}^{-1} = 2.26855 \times 10^{-18}\ \mathrm{s}^{-1}, \tag{1}$$

$$H_0^{-1} = 4.40811 \times 10^{17}\ \mathrm{s} = 13.97\ \mathrm{Gyr}. \tag{2}$$

(b) $c/H_0 = 4282.75\ \mathrm{Mpc} = 4.283\ \mathrm{Gpc}$. Since one parsec is 3.26156 light-years, this is $13.97\ \mathrm{Gly}$. The equality of the numerical values in Gyr and Gly is a consequence of using light-travel units, not a

new physical law.

(c) An object moving at constant speed has travel time D/v. If the expansion speed of a fixed pair had always been related to its separation through a constant H, then $D \propto e^{Ht}$ and there would be no finite time at which $D = 0$. The exact coasting example is the empty, negatively curved Milne model, $a \propto t$, for which $t_0 = H_0^{-1}$. The often-quoted "constant-speed since zero separation" picture gives the same scaling, but it is not a general solution of the Friedmann equation. In a matter-only flat model, for example, $t_0 = 2/(3H_0)$. **The age requires the history of H, not only its present value.**

(d)

$$\rho_{\mathrm{crit},0} = \frac{3H_0^2}{8\pi G} = 9.20 \times 10^{-27}\ \mathrm{kg\,m^{-3}}, \qquad \bar{\rho}_{m0} = 0.3\rho_{\mathrm{crit},0} = 2.76 \times 10^{-27}\ \mathrm{kg\,m^{-3}}. \tag{3}$$

Checks. H_0 has dimensions T^{-1}; multiplying it by a distance gives a speed. The mean matter density includes dark matter and is not a proton count.

Solution 2 — When does a galaxy trace the Hubble flow? Return to problem →

(a) $v_H = H_0 D = 140\ \mathrm{km\,s^{-1}}$. The peculiar contribution can be as large as $300/140 = 2.14$, or **214%** of the Hubble signal. The observed line-of-sight velocity could even have the opposite sign.

(b) Require $300/(H_0 D) \leq 0.05$. The most conservative choice is the smallest H_0 allowed in the question. Here H_0 is fixed at 70, so

$$D \geq \frac{300}{0.05 \times 70} = 85.7\ \mathrm{Mpc}. \tag{4}$$

(c) The measured velocity is approximately $H_0 D + v_{\mathrm{pec,los}}$. Andromeda's negative total line-of-sight shift is compatible with a negative local peculiar velocity in the Milky Way–Andromeda system. The smooth Hubble flow is not a rule that every nearby object must recede from every other nearby object.

(d) Independent random velocities partly cancel in an average, with a statistical error that typically decreases as the sample grows. A shared bulk flow is correlated: every galaxy receives a similar offset, so averaging does not remove it. One needs a model, an appropriate reference-frame correction and an estimate of correlated uncertainties.

Check. A fractional-error requirement must be applied to the signal, not to the sum after the unknown peculiar velocity has already been added.

Solution 3 — One redshift, three different questions. Return to problem →

(a) The wavelength ratio is exactly three, so $1 + z = 3$, $z = 2$, and $a_{\mathrm{em}} = 1/3$.

(b) Every source-frame time interval is stretched by $1+z$. The feature lasts **60 days** in the observer's frame.

(c) For a pure radial special-relativistic Doppler interpretation,

$$\beta = \frac{(1+z)^2-1}{(1+z)^2+1} = \frac{9-1}{9+1} = 0.800. \tag{5}$$

Thus the inferred local speed would be $0.800c$.

(d) That calculation compares two inertial frames in flat spacetime. A cosmological redshift compares two scale factors at separated emission and observation events. The proper-distance recession rate is $H(t)D(t)$ on a specified cosmic-time slice. It can be assigned today or at emission, and neither assignment is generally the same as the special-relativistic velocity constructed from z.

Checks. At low redshift all reasonable velocity definitions agree to first order. At $z=2$ they do not. The duration shift follows from the same crest-spacing argument as the wavelength shift.

Solution 4 — The ruler that did not stretch. Return to problem →

(a) With $a_0 = 1$, the comoving separation is $12{,}000\ \mathrm{Mpc}$. At $z=2$, $a = 1/3$, so

$$D_{\mathrm{em}} = aD_0 = 4000\ \mathrm{Mpc} = 4.00\ \mathrm{Gpc}. \tag{6}$$

(b) The proper separation grew by a factor of three. The comoving separation did not change.

(c) No. A bound group is not a pair of free comoving test particles. Its internal dynamics and local gravity determine its size; a metre ruler is held together by electromagnetic forces. Cosmological expansion is not a universal stretching command applied to every object.

(d) The benchmark gives $E(2) = \sqrt{0.3(3)^3 + 0.7} = \sqrt{8.8} = 2.966$. Hence

$$\frac{H(2)D_{\mathrm{em}}}{c} = \frac{70(2.966)(4000)}{299792.458} \simeq 2.77. \tag{7}$$

This is a rate of change of a large proper separation, not a local passage speed. Each galaxy can be locally at rest in its own comoving frame. No material object or signal overtakes a nearby light ray.

Check. The same result follows from $H_0 D_0 E(2)/(1+z)$. Keeping the factor $1+z$ in the denominator is essential: the separation at emission was smaller.

Solution 5 — Critical does not mean heavy. Return to problem →

(a) For zero curvature, and including Λ as one of the energy components, $H^2 = 8\pi G\rho_{\mathrm{tot}}/3$. Solving for the total mass-equivalent density gives the stated result. “Critical” refers to this total, not to matter alone.

(b) Multiply by c^2 with c in metres per second:

$$u_{\text{crit}} = 9.2039 \times 10^{-27}(2.99792458 \times 10^{8})^2 = 8.27 \times 10^{-10}\ \text{J m}^{-3}. \tag{8}$$

(c) $9.2039 \times 10^{-27}/1.67 \times 10^{-27} = 5.51$ proton masses per cubic metre.

(d) Much of the cosmic energy budget is dark energy and nonbaryonic dark matter. Even the baryonic matter is not all free protons. This is an equivalent mass-density comparison, not a census of particles.

Common unit trap. Using $c = 299792.458$ as though it were in metres per second makes the energy density wrong by 10^6.

Solution 6 — A thermodynamic derivation of three eras. Return to problem →

(a) Expand $d(uV) = -P\,dV$:

$$V\,du + (u+P)dV = 0, \qquad \frac{dV}{V} = 3\frac{da}{a}. \tag{9}$$

Therefore $du/u = -3(1+w)da/a$, giving

$$u(a) = u_0 a^{-3(1+w)}. \tag{10}$$

(b) Matter: $w = 0$, so $\rho_m \propto a^{-3}$. Radiation: $w = 1/3$, so $u_r \propto a^{-4}$. Λ: $w = -1$, so u_Λ is constant.

(c) At $1 + z = 4$, the ratios are **64, 256 and 1**, respectively.

(d) Photon number per comoving volume is fixed in the ideal freely streaming case, so number density contributes a^{-3}. Each photon's energy is proportional to frequency and therefore to a^{-1}. The product is a^{-4}.

(e) The acceleration source is $u + 3P = u(1+3w)$. For positive energy density, acceleration requires $w < -1/3$. At equality $w = -1/3$, the dominant component gives coasting expansion in a flat model.

Checks. Set $w = 0$: the total mass in a comoving volume is constant. Set $w = -1$: the energy density is constant, so the energy in an increasing physical volume increases. That is allowed in the gravitational setting and is not a recipe for a perpetual-motion machine.

Solution 7 — The CMB was once warm. Return to problem →

(a)

$$T_{\text{ls}} = 2.725(1091) = 2973\ \text{K}, \qquad \lambda_{\text{max}} = \frac{2.89777 \times 10^{-3}}{2973} = 9.75 \times 10^{-7}\ \text{m} = 0.975\ \mu\text{m}. \tag{11}$$

(b) Because $u_\gamma \propto T^4 \propto (1+z)^4$,

$$u_{\gamma,\mathrm{ls}} = 4.17 \times 10^{-14} (1091)^4 = 5.91 \times 10^{-2}\ \mathrm{J\,m^{-3}}. \tag{12}$$

(c) Every freely propagating photon's wavelength has increased by **1091**. Its individual energy has decreased by the same factor.

(d) The emitting plasma was hot. The present temperature is the result of subsequent expansion. The last-scattering surface labels where and when photons last scattered; it is not a cold material boundary surrounding us.

(e) B_ν and B_λ are densities per unit frequency and per unit wavelength. They are related by $|d\nu/d\lambda| = c/\lambda^2$, which changes the location of the maximum. The temperature is unique; the numerical wavelength at "the peak" depends on the representation.

Check. Multiplying $0.975\ \mu\mathrm{m}$ by 1091 gives approximately $1.064\ \mathrm{mm}$, the present B_λ peak.

Solution 8 — Equality is not acceleration. Return to problem →

(a) Equality requires $\Omega_{m0}(1+z)^3 = \Omega_{\Lambda 0}$, so

$$z_{m\Lambda} = (0.7/0.3)^{1/3} - 1 = 0.326. \tag{13}$$

(b) For Λ, $\rho_\Lambda + 3P_\Lambda/c^2 = -2\rho_\Lambda$. The total acceleration source vanishes when $\rho_m = 2\rho_\Lambda$:

$$z_{\mathrm{acc}} = (2 \times 0.7/0.3)^{1/3} - 1 = 0.671. \tag{14}$$

(c) The acceleration transition occurs earlier, at larger redshift. Λ need only reach half the matter density to cancel the matter term in the acceleration equation; it must reach the full matter density for equality.

(d) $q_0 = \Omega_{m0}/2 - \Omega_{\Lambda 0} = 0.15 - 0.70 = -0.55$.

(e) Yes. Acceleration means $\ddot{a} > 0$, or $q < 0$, which gives $\dot{H}/H^2 > -1$. Thus $\dot{H}$ can be negative while remaining less negative than $-H^2$. In the benchmark $\dot{H} = -H^2(1+q) = -0.45H^2 < 0$, while $\ddot{a}/a = 0.55H^2 > 0$.

Check. Do not identify a decreasing expansion rate H with a decreasing recession speed for every fixed comoving pair. Their separation is also growing.

Solution 9 — A standard candle and a standard ruler. Return to problem →

(a) At $z = 1$,

$$d_L = 2D_C = 6.60\ \mathrm{Gpc}, \qquad d_A = D_C/2 = 1.65\ \mathrm{Gpc}. \tag{15}$$

The emission-time proper distance is also $D_C/2 = 1.65\ \mathrm{Gpc}$ in this flat model. Equality of d_A and emission-time proper distance follows from the definition of d_A, not from an accidental numerical coincidence.

(b) Convert $6.60\ \mathrm{Gpc} = 2.037 \times 10^{26}\ \mathrm{m}$. Then

$$F = \frac{4.00 \times 10^{36}}{4\pi (2.037 \times 10^{26})^2} = 7.67 \times 10^{-18}\ \mathrm{W\,m^{-2}}. \tag{16}$$

(c)

$$\theta = \frac{20.0\ \mathrm{kpc}}{1.65 \times 10^{6}\ \mathrm{kpc}} = 1.212 \times 10^{-5}\ \mathrm{rad} = 2.50\ \mathrm{arcsec}. \tag{17}$$

(d) $d_L/d_A = 6.60/1.65 = 4 = (1+z)^2$.

(e) Examples: a finite observed filter samples a different rest-frame wavelength interval, requiring a K-correction; dust or source evolution can alter the inferred luminosity. Detector calibration, lensing and the source's spectral shape can also matter.

Check. The flux uses luminosity distance; the angle uses angular-diameter distance. Using one in place of the other produces a factor-of-four error in flux at this redshift.

Solution 10 — The age is an integral. Return to problem →

(a) For matter only, $H = H_0 a^{-3/2}$, so

$$t_0 = \int_0^1 \frac{da}{aH} = \frac{1}{H_0}\int_0^1 a^{-1/2}da = \frac{2}{3H_0} = 9.31\ \mathrm{Gyr}. \tag{18}$$

(b) For the flat matter-plus-Λ model,

$$t_0 = \frac{1}{H_0}\int_0^1 \frac{da}{\sqrt{0.3/a + 0.7a^2}} = \frac{2}{3H_0\sqrt{0.7}}\,\mathrm{arsinh}\sqrt{\frac{0.7}{0.3}}. \tag{19}$$

The result is **13.47 Gyr**. To obtain the closed form, substitute $u = a^{3/2}$; the integral becomes proportional to $\int du/\sqrt{\Omega_{m0} + \Omega_{\Lambda 0}u^2}$.

(c) At earlier times the matter-only model expands faster relative to its present rate: $E(a) = a^{-3/2}$. Adding Λ lowers the past $E(a)$ relative to the same H_0, so more time is required to reach a given scale factor. The age is the accumulated dt, not the reciprocal of the final rate.

(d) Radiation grows as a^{-4} and eventually dominates the early expansion. The real age integral should include it, along with the appropriate relativistic species. The correction to the total age is small compared with the difference between the two toy models, but it is not zero. Precision parameters also have uncertainties.

Checks. The integrand behaves as $a^{-1/2}$ near zero in the matter model, so the integral converges. Setting Λ to zero in the analytic expression must recover $2/(3H_0)$; use the limit rather than substituting into a formula containing $1/\sqrt{\Omega_\Lambda}$ blindly.

Solution 11 — A circle inside curved space. Return to problem →

(a) A geodesic radius r corresponds to polar angle $\alpha = r/R$ on the sphere. The circle's embedding-space radius is $R\sin\alpha$, so

$$C = 2\pi R \sin(r/R). \tag{20}$$

(b) Using $\sin x = x - x^3/6 + \cdots$,

$$C = 2\pi r - \frac{\pi r^3}{3R^2} + \cdots, \qquad \frac{2\pi r - C}{2\pi r} \simeq \frac{r^2}{6R^2}. \tag{21}$$

(c) Set the absolute deficit to $1\ \mathrm{m}$:

$$r \simeq (3R^2 \times 1\ \mathrm{m})^{1/3} = [3(6.4\times10^6\ \mathrm{m})^2]^{1/3} = 4.97\times10^4\ \mathrm{m} \approx 49.7\ \mathrm{km}. \tag{22}$$

This is an ideal one-metre total circumference difference. A real experiment would need a significance criterion and an error budget, not merely a nonzero predicted difference.

(d) The correction is small because $r/R \ll 1$, not because it vanishes. With better precision, smaller circles can reveal curvature. Conversely, a null result places a lower bound on the curvature radius within the assumed geometry.

Check. At the north pole, a “circle of radius one metre” follows the surface, not a straight chord through the Earth. Confusing those radii changes the formula.

Solution 12 — Three horizons in one model. Return to problem →

Idea. A horizon is a cumulative light-travel question. Integrate along a null path; do not guess from the current Hubble radius.

(a) In EdS, $a = (t/t_0)^{2/3}$. Thus

$$D_{\rm ph}(t) = a(t)c\int_0^t \frac{dt'}{a(t')} = ct^{2/3}\int_0^t (t')^{-2/3}dt' = 3ct. \tag{23}$$

Since $H = 2/(3t)$, $D_{\rm ph} = 2c/H$.

(b) The future conformal-time integral is

$$\int_t^\infty \frac{dt'}{a(t')} \propto \int_t^\infty (t')^{-2/3}dt' = \infty. \tag{24}$$

There is **no finite future event horizon** in this ideal model. Any finite comoving separation can eventually be crossed by light.

(c) In pure de Sitter, $a = e^{H(t-t_0)}$, and

$$D_{\rm eh,0} = c\int_{t_0}^\infty e^{-H(t-t_0)}dt = \frac{c}{H}. \tag{25}$$

The equality with the Hubble radius is special to this model.

(d) The particle horizon is a present proper distance obtained by stretching all the small comoving steps traversed by the photon. Those steps were made when the scale factor was smaller. Their present total can exceed both the distance light would travel in a static space during the age and the instantaneous length c/H_0. No local light speed exceeds c.

Checks. The EdS particle horizon grows as t, while c/H grows as t with a different coefficient. In de Sitter, c/H stays fixed while the comoving event horizon shrinks.

Website-sourced problems: worked solutions

Solution 13 — When do distance definitions matter? Return to problem →

Source. Sheffield PHY306, Problem 4. The following is an independent worked adaptation.

(a) Distance duality gives $d_L/d_A = (1+z)^2$. Relative to d_A, the fractional difference is $(1+z)^2 - 1$. Setting this equal to 0.05 gives

$$z = \sqrt{1.05} - 1 = 0.0247 \approx 0.025. \tag{26}$$

The leading estimate $2z = 0.05$ gives the same result to this accuracy.

(b) Require $v_{\rm pec}/(H_0 D) < 0.1$. The smallest H_0 gives the largest required distance:

$$D > \frac{1000}{0.1 \times 50} = 200 \text{ Mpc}. \tag{27}$$

(c) At $D = 200$ Mpc, $z \simeq H_0 D/c$ gives 0.0334 for $H_0 = 50$ and 0.0667 for $H_0 = 100$. The corresponding leading distance-definition differences are about 6.7% and 13%, both larger than 5%.

(d) "Matters" depends on the target precision, the observable and the acceptable systematic error. Also, d_L and d_A are not two noisy measurements of one distance; they answer different operational questions. The 5% comparison is a useful scale estimate, not a law that says one definition may be ignored below a particular redshift.

Check. The two parts identify different systematics: confusing definitions and contaminating velocities. Solving one does not remove the other.

Solution 14 — Count the photons, not just their energies. Return to problem →

Source. Sheffield PHY306, Problem 3, rebuilt as a photon-counting argument.

(a) The received photon energy is $N\bar{E}_{\rm em}/(1+z)$. The arrival interval is $\Delta t_{\rm obs} = (1+z)\Delta t_{\rm em}$. Divide by area and time:

$$F = \frac{N\bar{E}_{\rm em}}{4\pi D_C^2 (1+z)^2 \Delta t_{\rm em}} = \frac{L}{4\pi[(1+z)D_C]^2}. \tag{28}$$

Comparison with $F = L/(4\pi d_L^2)$ gives $d_L = (1+z)D_C$.

(b) At emission, a small angle $d\theta$ subtends proper length $\ell = a_{\rm em} D_C d\theta$ in flat space. Thus $d_A = \ell/d\theta = a_{\rm em} D_C = D_C/(1+z)$.

(c) Eliminating D_C gives $d_L = (1+z)^2 d_A$.

(d) Photon conservation enters when the received number is taken to be the emitted number. If photons disappear, the flux acquires an additional transmission factor. Spatial flatness is a geometric statement; it does not guarantee that photons cannot be absorbed or converted. Distance duality assumes the relevant photon conservation and metric propagation conditions.

Check. Each factor of $1+z$ has a separate physical origin. Remembering "two factors" without knowing energy versus arrival rate makes it easy to lose one in a new problem.

Solution 15 — The Einstein–de Sitter toolkit. Return to problem →

Source. Yale ASTR 610, Problem Set 1, Problem 1.

(a) The Friedmann equation gives $\dot{a}/a = H_0 (a/a_0)^{-3/2}$. Let $A = a/a_0$. Then $A^{1/2} dA = H_0 dt$, so $(2/3)A^{3/2} = H_0 t$ after choosing $A = 0$ at $t = 0$. Therefore $A = (3H_0 t/2)^{2/3}$.

(b) Matter conservation gives $\rho = \rho_{\mathrm{crit},0} A^{-3}$. Since $A^3 = (3H_0 t/2)^2$,

$$\rho = \frac{3H_0^2}{8\pi G}\frac{4}{9H_0^2 t^2} = \frac{1}{6\pi G t^2}. \tag{29}$$

Also $H = \dot{A}/A = 2/(3t)$, so $Ht = 2/3$.

(c) The particle horizon is $D_{\mathrm{ph}} = ac\int_0^t dt'/a = 3ct = 2c/H$, as derived in Solution 12.

(d) At $H_0 = 70$, $t_0 = 9.31$ Gyr and $D_{\mathrm{ph}} = 27.94$ Gly $= 8.57$ Gpc. The horizon exceeds $ct_0 = 9.31$ Gly because it is a present proper distance, accumulated while the universe expanded.

Checks. The density result contains no H_0: in this model, specifying the cosmic age fixes the density. That is a strong dimensional and physical consistency check.

Solution 16 — Can tired light pass every test? Return to problem →

Source. Yale ASTR 610, Problem 3, extended with duration and spectral checks.

(a) Integration gives $E(r) = E_0 e^{-Kr}$. Since photon energy is inversely proportional to wavelength,

$$1 + z = \frac{E_0}{E(r)} = e^{Kr}, \qquad z = e^{Kr} - 1. \tag{30}$$

(b) For $Kr \ll 1$, $z \simeq Kr$. Matching $cz \simeq H_0 r$ gives $K = H_0/c$, hence

$$K = \frac{70}{299792.458} = 2.33 \times 10^{-4}\ \mathrm{Mpc}^{-1}. \tag{31}$$

(c) The stipulated toy model predicts **20 days**, because it does not change clock rates. Expansion predicts **40 days** at $z = 1$. Thus matching a low-redshift distance–redshift line is not enough to make the models equivalent.

(d) A blackbody is not specified by a shifted wavelength alone. Its photon occupation, spectral intensity and temperature normalization must also match. Free expansion preserves the Planck form with $T \propto a^{-1}$ and produces the associated surface-brightness dimming. The simple tired-light rule gives no mechanism for all of these relations. A rival model would have to specify how photon number, arrival rates, angular propagation and spectral intensity transform, and then pass the corresponding observations.

Check. We have ruled out the particular toy model under the stated assumptions, not proved that no imaginable alternative theory could ever be constructed. Scientific comparisons must specify the model being tested.

Solution 17 — Equality in a different thermal universe. Return to problem →

Source. Yale ASTR 610, Problem 2. The unusually hot photon background and permanently relativistic neutrinos are deliberate hypothetical inputs.

(a) Matter scales as $(1+z)^3$; both relativistic components scale as $(1+z)^4$. Equality gives $1+z_{\rm eq} = \Omega_{m0}/\Omega_{r0}$, with $\Omega_{r0} = \Omega_{\gamma 0} + \Omega_{\nu 0} = 2.5\Omega_{\gamma 0}$.

(b) Using $u_\gamma = 4\sigma T^4/c$ and $\rho_{\rm crit} = 3(100h)^2/(8\pi G)$ in consistent units gives

$$\Omega_{\gamma 0} h^2 = 4.482 \times 10^{-3} \quad (T_0 = 10.0\ \mathrm{K}). \tag{32}$$

This is much larger than the photon contribution at 2.725 K because radiation density scales as T^4.

(c)

$$1 + z_{\rm eq} = \frac{\Omega_{m0} h^2}{2.5(4.482 \times 10^{-3})} = 89.3\, \Omega_{m0} h^2. \tag{33}$$

The question asks for the coefficient in terms of $\Omega_{m0}h^2$; no numerical matter density was supplied.

(d) A relativistic gas has $P = u/3$ and its energy density falls as a^{-4}. Nonrelativistic neutrinos would instead contribute approximately to matter, changing both the equality condition and the expansion history. “Neutrino” names a particle, not a permanent equation of state.

Check. Do not import the real universe's equality redshift or neutrino temperature ratio into a problem that explicitly replaces the thermal assumptions.

Solution 18 — Why flatness needs an explanation. Return to problem →

Source. MIT 8.286, Quiz 3 Review, Problem 9. Sign conventions follow Section 4.

(a) Divide the Friedmann equation by H^2 and move the curvature term to the other side:

$$\Omega_{\rm tot} - 1 = -\Omega_k = \frac{kc^2}{a^2 H^2}. \tag{34}$$

(b) Since $H^2\Omega_{\rm tot} = 8\pi G\rho/3$,

$$\frac{\Omega_{\rm tot} - 1}{\Omega_{\rm tot}} = \frac{3kc^2}{8\pi G \rho a^2}. \tag{35}$$

For matter, $\rho a^2 \propto a^{-1}$, so the ratio is proportional to $a \propto t^{2/3}$.

(c) For radiation, $\rho a^2 \propto a^{-2}$, so the ratio is proportional to $a^2 \propto t$.

(d) With constant H and $a \propto e^{N}$,

$$|\Omega_{\rm tot} - 1| \propto e^{-2N}. \tag{36}$$

Inflation drives the observable universe very close to flatness by increasing aH. It does not change the constant curvature sign or the underlying spatial identifications; it makes the curvature radius enormous compared with the scales we probe.

Check. During decelerating expansion the comoving Hubble radius grows and the flatness deviation grows. During inflation it shrinks and the deviation is suppressed. The same geometric quantity explains both statements.

Solution 19 — A very distant quasar. Return to problem →

Source. MIT 8.286, Quiz 3 Review, Problem 10. The historical redshift is used as a mathematical example; no "most distant object" status is claimed today.

Idea. Keep four quantities separate: age now, age then, present proper distance and emission-time proper distance. They are related, but they are not interchangeable.

(a) Let $E(z) = \sqrt{0.35(1+z)^3 + 0.65}$. Then

$$t_0 = \frac{1}{H_0}\int_0^{\infty} \frac{dz}{(1+z)E(z)}, \tag{37}$$

$$t_e = \frac{1}{H_0}\int_{5.82}^{\infty} \frac{dz}{(1+z)E(z)}, \qquad D_0 = \frac{c}{H_0}\int_0^{5.82} \frac{dz}{E(z)}, \qquad D_e = \frac{D_0}{6.82}. \tag{38}$$

Equivalently, integrate t_e from $a = 1/6.82$ to zero using $da/(aH)$. The upper limit infinity is a compact way to include all earlier times in this idealized model.

(b) The present proper-distance recession rate is $v_{\rm rec,0} = H_0 D_0$. It uses the present separation and the present rate.

(c) Numerical evaluation gives

$$t_0 \simeq 13.9\ \text{Gyr}, \qquad t_e \simeq 0.95\ \text{Gyr}, \tag{39}$$

$$D_0 \simeq 27.2\ \text{Gly}, \qquad D_e \simeq 3.99\ \text{Gly}. \tag{40}$$

The present recession rate is about $1.81c$ with the stated definition. The lookback time is $t_0 - t_e \simeq 12.9\ \text{Gyr}$.

(d) During the photon's journey the source and observer became farther apart. The photon traversed a sequence of local intervals at c while the distance scale changed. A present proper distance is a snapshot quantity; it is not the distance the photon travelled through a static space.

Checks. $D_e(1+z) = D_0$. The emission age is positive and smaller than the present age. The superluminal proper-distance rate is not a locally measured velocity.

Solution 20 — A BAAO calculation and its approximation. Return to problem →

Source. BAAO 2015 Round 2, Question 12. Parts (a)–(c) retain its deliberately simple low-redshift model; parts (d)–(e) expose the approximation.

(a) $\lambda_{\text{obs}} = 486.1(1.30) = 631.93$ nm, or **632 nm**.

(b) $D = cz/H_0 = 299792.458(0.30)/72 = 1249$ Mpc, approximately **1.25 Gpc**.

(c) In the same linear model, $t = D/c = z/H_0 = 4.07$ Gyr.

(d) In the specified flat $(0.3, 0.7)$ model with $H_0 = 72$, numerical integration gives approximately

$$t_{\text{LB}}(0.30) = 3.32 \text{ Gyr}, \qquad d_L(0.30) = 1.51 \text{ Gpc}. \tag{41}$$

The linear law replaces a curved distance–redshift relation by its tangent and also fails to distinguish distance definitions. At $z = 0.30$, those differences are already appreciable. The original short question is a useful estimate exercise, not a precision cosmological inference.

(e) The special-relativistic formula gives $\beta = (1.3^2 - 1)/(1.3^2 + 1) = 0.2565$. Treating that as the present HD_0/c would mix a local inertial-frame Doppler construction with a cosmological proper-distance rate.

Check. We have not "corrected the BAAO answer" by changing its assumptions. We solved the stated approximation first, then compared it with a richer model. That distinction is essential when using past papers.

Original Olympiad challenges: worked solutions and reasoning

How to read these solutions. Problems 21–30 are **new MAIS PAC exercises**, not official IOAA questions. Each identifies the past-material theme that inspired it. The "Idea" paragraph explains how to recognize the governing physics; the derivation supplies the calculation; the "Checks" paragraph tests the result. The solutions are independent and are not official IOAA marking schemes.

Solution 21 — A rocket against the Hubble flow. Return to problem →

IOAA connection. IOAA 2022 Theory 4 studies travel between galaxies while the background expands. This version adds a threshold, a new travel-time calculation and a limiting-case discussion.

Idea. The rocket's peculiar speed is measured locally, but the distance to the destination also grows. Write an equation for the **remaining proper distance**, rather than subtracting a constantly changing recession speed by hand. The supplied differential equation already contains both effects.

(a) Solve $\dot{D} - HD = -u$ using the integrating factor e^{-Ht}:

$$\frac{d}{dt}\left(De^{-Ht}\right) = -ue^{-Ht}. \tag{42}$$

Integrating from zero to t gives

$$D(t) = \frac{u}{H} + \left(D_0 - \frac{u}{H}\right)e^{Ht}. \tag{43}$$

The destination itself is at $D_{\text{dest}} = D_0 e^{Ht}$. The rocket's actual proper position is $D_{\text{dest}} - D(t)$; $D(t)$ in the solution is the remaining separation.

(b) If $u > HD_0$, the coefficient of e^{Ht} is negative and the remaining separation eventually reaches zero. If $u < HD_0$, the expression instead reaches a minimum and then grows without bound. The threshold is

$$u_{\text{th}} = HD_0 = 70 \times 10 = 700\ \text{km s}^{-1}. \tag{44}$$

A 1000 km s^{-1} rocket **can** arrive under the idealized assumptions.

(c) Set $D(t) = 0$:

$$e^{Ht} = \frac{u}{u - HD_0}, \qquad t = \frac{1}{H}\ln\left(\frac{u}{u - HD_0}\right). \tag{45}$$

For $u = 1500$,

$$t = \frac{1}{H}\ln\left(\frac{1500}{800}\right) = 0.6286H^{-1} = 8.78\ \text{Gyr}. \tag{46}$$

(d) As $u \to (HD_0)^+$, the denominator approaches zero and the arrival time diverges. At equality the remaining distance approaches D_0 asymptotically rather than reaching zero. Just exceeding the threshold is not enough for a practically short journey.

(e) $D_0/u = 6.98$ Gyr ignores the increasing separation. It is recovered only in the limit $H \to 0$. Constant H is an exact de Sitter-like idealization, whereas the real expansion rate changes with the

cosmic contents; local gravity, fuel and relativistic trajectory details are also neglected.

Checks. For $u \gg HD_0$, expand the logarithm to obtain $t \simeq D_0/u$. At $u = HD_0$, the formula correctly signals no finite arrival time. A negative time would indicate that the wrong branch or the wrong sign was used.

Why this is hard. The phrase "constant speed" refers to a local peculiar speed, not a constant rate of change of the remaining distance. Separating those two ideas is the whole problem.

Solution 22 — Weighing dark energy, then asking when it wins. Return to problem →

IOAA connection. The density-budget and equality calculation is inspired by IOAA 2023 Theory 5. The pressure and acceleration parts are new conceptual extensions.

Idea. Flatness fixes the total density. Subtract matter to find Λ, then use the different redshift scalings. Equality compares densities; acceleration compares the gravitational source $\rho + 3P/c^2$.

(a) The critical density is $9.2039 \times 10^{-27}\ \mathrm{kg\,m^{-3}}$. Therefore

$$\rho_\Lambda = \rho_{\mathrm{crit}} - \rho_{m0} = 6.404 \times 10^{-27}\ \mathrm{kg\,m^{-3}}, \tag{47}$$

$$u_\Lambda = \rho_\Lambda c^2 = 5.76 \times 10^{-10}\ \mathrm{J\,m^{-3}}. \tag{48}$$

(b) $\Omega_{m0} = 2.8/9.2039 = 0.304$, and $\Omega_{\Lambda 0} = 0.696$. These are close to, but not exactly, the rounded $(0.3, 0.7)$ benchmark.

(c) Equality:

$$1 + z_{m\Lambda} = (\rho_\Lambda/\rho_{m0})^{1/3} = 1.318, \qquad z_{m\Lambda} = 0.318. \tag{49}$$

Acceleration begins when $\rho_m = 2\rho_\Lambda$, so

$$1 + z_{\mathrm{acc}} = (2\rho_\Lambda/\rho_{m0})^{1/3} = 1.660, \qquad z_{\mathrm{acc}} = 0.660. \tag{50}$$

(d)

$$q_0 = \frac{1}{2}\Omega_{m0} - \Omega_{\Lambda 0} = -0.544. \tag{51}$$

(e) A homogeneous cosmological component has no spatial pressure gradient, so there is no ordinary pressure-gradient force of the kind the student imagines. In general relativity, pressure itself contributes to the gravitational source. For Λ, $\rho_\Lambda + 3P_\Lambda/c^2 = -2\rho_\Lambda$, giving an accelerating contribution.

Checks. The two densities sum to the critical density. The acceleration redshift exceeds the equality redshift by the factor $2^{1/3}$ in $1+z$. The negative value of q_0 agrees with the stated accelerating universe.

Why this is hard. The same symbol ρ appears in two different equations with different weights. Keeping the bookkeeping explicit prevents the common “equal densities means acceleration starts” error.

Solution 23 — A warm spectrum and a deliberately wrong clock. Return to problem →

IOAA connection. IOAA 2024 Theory 5 combines a CMB wavelength shift with an explicitly matter-dominated age estimate. This version uses a different wavelength and asks why the assumption matters.

Idea. The spectrum gives a temperature ratio, hence a scale-factor ratio. The age requires a separate cosmological model. Do not let a successful temperature calculation smuggle in an unjustified expansion history.

(a)

$$T = \frac{2.898 \times 10^{-3}}{5.00 \times 10^{-4}} = 5.796\ \mathrm{K}, \tag{52}$$

$$1 + z = \frac{T}{T_0} = 2.1268, \qquad z = 1.127, \qquad a = 0.470. \tag{53}$$

(b) In the stipulated matter-only model, $H = H_0(1+z)^{3/2}$ and

$$t(z) = \frac{2}{3H_0}(1+z)^{-3/2} = 9.312(2.1268)^{-3/2}\ \mathrm{Gyr} \simeq 3.00\ \mathrm{Gyr}. \tag{54}$$

(c) At $z \simeq 1.13$, Λ is dynamically important in the benchmark model. Ignoring it changes the inferred age substantially: the flat $(0.3, 0.7)$ model with the same H_0 gives about **5.28 Gyr** at this redshift. Radiation is not the dominant missing ingredient here; Λ is. At much higher redshift, radiation would also need to be included.

(d) A pure radial Doppler interpretation gives

$$\beta = \frac{(2.1268)^2 - 1}{(2.1268)^2 + 1} = 0.638. \tag{55}$$

The CMB's near-isotropy, blackbody spectrum, relation to the hot early universe and consistency with other cosmological probes support cosmological expansion, not a shell of material moving away from us at that local speed. Our actual peculiar motion produces a much smaller dipole pattern.

Checks. The temperature is only about twice today's value, so this is a low-redshift CMB exercise, not last scattering. The age is smaller than the model's present age. The matter-only answer is correct **for the supplied toy model**, even though it is not a realistic precision estimate.

Why this is hard. Olympiad questions sometimes impose an approximation that is not physically exact. The right response is to use it where requested, then identify its domain of validity—not silently replace the question with a different model.

Solution 24 — A small scar on a blackbody. Return to problem →

IOAA connection. IOAA 2020 Team 14 investigates CMB spectral distortions. This exercise isolates one small piece of that physics: an energy budget at fixed photon number. It does not reproduce the official problem or model the full reaction network.

Idea. A chemical potential changes both number and energy at fixed temperature. But the problem fixes photon number, so the temperature must adjust. Include that adjustment before relating the injected energy to μ.

(a) Let $\tau = \Delta T/T_i$. To first order,

$$\frac{N_f}{N_i} \simeq 1 + 3\tau - 1.368\mu. \tag{56}$$

Fixed photon number gives $\tau = (1.368/3)\mu = 0.456\mu$. For energy,

$$\frac{u_f}{u_i} \simeq 1 + 4\tau - 1.111\mu. \tag{57}$$

Therefore the injected fractional energy is

$$\epsilon = (4 \times 0.456 - 1.111)\mu = 0.713\mu, \qquad \boxed{\mu \simeq 1.40\epsilon}. \tag{58}$$

The coefficients 1.368 and 1.111 are the small-μ expansions of the photon-number and energy integrals; they were supplied so that no advanced integration is required.

(b) For $\epsilon = 10^{-5}$,

$$\mu \simeq 1.40 \times 10^{-5}, \qquad \frac{\Delta T}{T_i} = 0.456\mu \simeq 6.40 \times 10^{-6}. \tag{59}$$

The final temperature is slightly higher, while the positive chemical potential suppresses the low-frequency occupation relative to a blackbody at that final temperature.

(c) Holding N fixed determines the temperature adjustment. Holding T fixed as well would leave no freedom to satisfy both conservation and the added energy. The three requirements would generally be incompatible.

(d) If photon-producing reactions are efficient, the chemical potential can relax toward zero by creating photons. The final spectrum may remain nearly Planckian despite energy injection. The relevant outcome

depends on the competition between injection and thermalization rates, so a fixed-N estimate is not valid at arbitrary early times.

Checks. Positive injected energy gives positive μ and a positive temperature adjustment in this model. The result is first order in tiny quantities; products such as $\mu\tau$ have been neglected.

Why this is hard. The tempting one-line answer uses the energy distortion at fixed temperature and forgets the constraint. Writing the two linearized conservation equations makes the hidden degree of freedom visible.

Solution 25 — Counting neutrons into helium. Return to problem →

IOAA connection. The topic is inspired by IOAA 2025 Theory 10 on BBN. The numerical delay and assumptions here define a new, standalone counting problem.

Idea. Follow neutrons until they are locked into nuclei. A helium-4 nucleus contains two neutrons and two protons, so a neutron fraction is not the same as a helium mass fraction.

(a) With only radioactive decay during the delay,

$$r = r_0 e^{-\Delta t/\tau_n} = \frac{1}{6} e^{-200/880} = 0.1328. \tag{60}$$

(b) Take $n_p = 1$ and $n_n = r$. All neutrons form $n_{\mathrm{He}} = r/2$ helium nuclei, consuming r protons. The remaining hydrogen count is $1 - r$. The helium mass fraction is

$$Y_p = \frac{4(r/2)}{(1-r) + 4(r/2)} = \frac{2r}{1+r} = 0.234. \tag{61}$$

Thus about **23.4%** of the baryonic mass is helium-4 in this simplified model.

(c)

$$\frac{n_{\mathrm{He}}}{n_{\mathrm{H}}} = \frac{r/2}{1-r} = \frac{r}{2(1-r)} = 0.0764. \tag{62}$$

Equivalently, there is roughly one helium nucleus per 13.1 hydrogen nuclei. The intermediate ratio $r/2 = 0.0664$ is helium nuclei per **initial** proton count, not per remaining hydrogen nucleus; the distinction matters.

(d) $r/(1+r)$ is the fraction of baryon **number** in neutrons. Helium mass counts four nucleons per nucleus, including the protons bound with those neutrons. Counting nuclei, nucleons and mass are three different operations.

(e) A higher baryon density generally lets deuterium survive and nucleosynthesis proceed earlier, leaving less time for neutron decay and tending to increase the helium yield. The real calculation also includes

expansion, weak freeze-out, nuclear reaction rates, the photon-to-baryon ratio and the destruction and production of several light nuclides.

Checks. As $r \to 0$, $Y_p \to 0$. If $r = 1$, all nucleons can be placed in helium and $Y_p = 1$. The answer lies between zero and one, as a mass fraction must.

Why this is hard. The physics is mostly bookkeeping, but the bookkeeping changes when neutrons become bound. Drawing a small ledger of protons, neutrons, helium nuclei and hydrogen nuclei is more reliable than memorizing a fraction.

Solution 26 — A hidden magnitude error. Return to problem →

IOAA connection. This original data exercise develops the distance-ladder and standard-candle reasoning that appears across IOAA materials. It is not a reproduction of a particular official data-analysis question.

Idea. A magnitude error is a logarithmic distance error. Propagate it through the distance modulus, then through the low-redshift relation $H_0 \simeq cz/d$.

(a) The inferred distance modulus is too large by 0.050 mag, because $\mu = m - M$ and the assumed M is more negative. Since $\mu = 5\log_{10}(d_L/10\ \mathrm{pc})$,

$$\frac{d_{\mathrm{assumed}}}{d_{\mathrm{true}}} = 10^{+0.050/5} = 10^{+0.01} = 1.02329. \tag{63}$$

Distances are overestimated by **2.33%**. An assumed candle that is intrinsically brighter than reality must be placed farther away to explain a given flux.

(b) At fixed measured redshift, $H_0 \propto 1/d$, so

$$\frac{H_{\mathrm{assumed}}}{H_{\mathrm{true}}} = 10^{-0.01} = 0.97724. \tag{64}$$

The Hubble constant is underestimated by **2.28%**. The two percentages differ slightly because they use different denominators.

(c)

$$H_{\mathrm{corrected}} = 70.0 \times 10^{+0.01} = 71.6\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{65}$$

(d) A constant magnitude offset multiplies all inferred distances by the same factor. It changes the normalization but not the shape of the relative distance–redshift relation. Acceleration is inferred from curvature in that relation, so the shape can remain informative after marginalizing over the unknown absolute calibration.

(e) No. A redshift-dependent error changes the shape and can mimic or hide acceleration. That is why population evolution, dust, selection and filter corrections are not interchangeable with a single harmless zero-point offset.

Checks. The sign passes a physical test: brighter assumed candles imply larger distances and a smaller inferred H_0. The correction raises, rather than lowers, the biased value.

Why this is hard. It is easy to lose the sign in a logarithm. Reconstructing the physical candle argument provides an independent check on the algebra.

Solution 27 — One ruler, two expansion measurements. Return to problem →

IOAA connection. This original problem combines angular, radial and distance-duality reasoning in the style of IOAA multi-part questions. No official BAO dataset is reproduced.

Idea. The same comoving length is viewed across the sky in one measurement and along the line of sight in the other. The transverse observation gives a distance; the radial observation gives an expansion rate.

(a)

$$D_M = \frac{r_d}{\theta} = \frac{147}{0.0300} = 4900 \text{ Mpc} = 4.90 \text{ Gpc}. \tag{66}$$

(b) A radial proper separation at that epoch is $r_d/(1+z)$, and $dD \simeq c\,dz/H(z)$. Equivalently, the comoving interval is $c\,dz/H$. Thus

$$H(z) = \frac{c\Delta z}{r_d} = \frac{299792.458(0.0450)}{147} = 91.8 \text{ km s}^{-1}\text{ Mpc}^{-1}. \tag{67}$$

(c) At $1+z = 1.5$,

$$d_A = \frac{4900}{1.5} = 3267 \text{ Mpc}, \qquad d_L = 4900(1.5) = 7350 \text{ Mpc}. \tag{68}$$

(d) The transverse distance is proportional to the assumed ruler, $D_M = r_d/\theta$, so a 2% increase in r_d increases D_M by 2%. The radial result is inversely proportional to it, $H = c\Delta z/r_d$, so the same increase decreases H by the factor $1/1.02$ (about 1.96%). The measured angle and redshift interval themselves do not change.

(e) Different dark-energy models can produce similar distances and expansion rates over a limited redshift range, especially after other parameters are adjusted. One needs a broader redshift baseline, complementary probes and a growth/geometry comparison to distinguish explanations. A ruler measurement does not directly identify a microscopic field or vacuum mechanism.

Checks. $H(0.5)$ is larger than the benchmark H_0, as expected. $d_A < D_M < d_L$ at positive redshift. The radial result would be absurdly small if the supplied interval were off by a large factor; checking the

expected scale is part of the solution.

Why this is hard. The factor $1+z$ is already accounted for when r_d is treated as a comoving ruler. Adding it again is a common error. Start from a proper or comoving interval and stay in that convention.

Solution 28 — Designing a curvature experiment. Return to problem →

IOAA connection. An original geometric-estimation problem in the Olympiad tradition, extending the Yale curved-space exercise. It tests whether a proposed measurement is actually sensitive enough.

Idea. Curvature appears through the dimensionless ratio r/R_c. Use the first correction to the circumference, then compare the predicted signal with the stated measurement precision.

(a) For a positively curved space,

$$C \simeq 2\pi r\left(1-\frac{r^2}{6R_c^2}\right). \tag{69}$$

The deficit is

$$\Delta C=\frac{\pi r^3}{3R_c^2}=\frac{\pi(2.00)^3}{3(20.0)^2}\ \text{Gpc}=0.0209\ \text{Gpc}=20.9\ \text{Mpc}. \tag{70}$$

(b)

$$\frac{\Delta C}{2\pi r}=\frac{r^2}{6R_c^2}=\frac{4}{2400}=1.67\times10^{-3}=0.167\%. \tag{71}$$

(c) No. A 1% fractional measurement is much coarser than a 0.167% signal. The enormous absolute deficit in megaparsecs is irrelevant if the circumference itself cannot be measured accurately enough.

(d) Examples: constructing a common cosmic-time slice across gigaparsecs; measuring geodesic distances rather than observables such as redshift and angle; peculiar velocities, evolution, lensing and selection effects; and the limited number of independent large-scale modes. Any real analysis must specify a model connecting these observables to spatial curvature.

(e) A small or undetected curvature could mean a curvature radius much larger than the surveyed region. It could also be compatible with different global topologies. Local geometry constrains, but does not by itself determine, the universe's total size or topology.

Checks. The fractional signal scales as $(r/R_c)^2$. Doubling the survey radius quadruples the leading sensitivity. The expansion of the sine is valid here because $r/R_c=0.1$.

Why this is hard. The problem asks for an experimental conclusion, not only a formula. Comparing signal to fractional uncertainty is the decisive step.

Solution 29 — The last message that can arrive. Return to problem →

IOAA connection. This new problem turns the expanding-background reasoning of IOAA 2022 Theory 4 into a null-propagation and event-horizon question.

Idea. For an incoming photon, the proper distance changes through two terms: expansion increases it at HD, while light reduces it at c. Solve $\dot{D} = HD - c$ with the initial condition, then ask whether the solution ever reaches zero.

(a) The solution is

$$D(t) = \frac{c}{H} + \left(D_0 - \frac{c}{H}\right) e^{Ht}. \tag{72}$$

(b) A finite arrival requires $D_0 < c/H$. The limiting present proper distance is

$$D_{\text{eh}} = \frac{c}{H} = 4.283\ \text{Gpc} = 13.97\ \text{Gly}. \tag{73}$$

This is the event horizon for signals emitted **now** in the stipulated constant-H future.

(c) Setting $D(t) = 0$ gives

$$t_{\text{arr}} = -\frac{1}{H} \ln\left(1 - \frac{HD_0}{c}\right). \tag{74}$$

For $D_0 = 0.8c/H$,

$$t_{\text{arr}} = \frac{\ln 5}{H} = 1.609 H^{-1} = 22.48\ \text{Gyr}. \tag{75}$$

(d) At exactly $D_0 = c/H$, $D(t) = c/H$ for all finite t. The photon remains at the horizon and never reaches us at a finite arrival time. It is the limiting boundary between arriving and non-arriving signals.

(e) The particle horizon concerns signals emitted in the past that arrive now. The event horizon concerns signals emitted now that may arrive in the future. Objects whose old light we see today can now be much farther away than the current event horizon; seeing their past does not mean a message sent now will reach us.

Checks. As $D_0 \to 0$, $t_{\text{arr}} \simeq D_0/c$. As $D_0 \to (c/H)^-$, the arrival time diverges. Both limits agree with physical intuition.

Why this is hard. “Can we see it?” and “can it signal us now?” refer to different emission events and different integrals. A spacetime sketch prevents the two questions from being merged.

Solution 30 — How much inflation is enough? Return to problem →

IOAA connection. An original enrichment problem inspired by early-universe themes in IOAA materials and the cited Sheffield/MIT exercises. The later growth factor is deliberately supplied, so no hidden thermal-history model is required.

Idea. During inflation, H is nearly constant while a grows exponentially. The curvature deviation is proportional to $(aH)^{-2}$, so it is exponentially suppressed. After inflation, ordinary decelerating evolution partly restores that deviation.

(a)

$$\frac{|\Omega_{\rm tot}-1|_{\rm end}}{|\Omega_{\rm tot}-1|_{\rm start}} = e^{-2N}. \tag{76}$$

(b) Starting from order unity and requiring a final value below 10^{-5} immediately after inflation gives

$$e^{-2N} < 10^{-5} \quad \Rightarrow \quad N > \frac{1}{2}\ln(10^5) = 11.51. \tag{77}$$

Thus **12 e-folds** suffice for this isolated flatness requirement if an integer count is requested.

(c) The later evolution multiplies the deviation by 10^{26}. The present requirement is

$$10^{26}e^{-2N} < 10^{-5} \quad \Rightarrow \quad N > \frac{1}{2}\ln(10^{31}) = 71.38. \tag{78}$$

Thus **72 e-folds** are required under the supplied estimate.

(d) The first bound ignores the long later interval during which the comoving Hubble radius grows. The second includes it. The actual required number depends on the inflationary energy scale, the reheating history, the expansion history and the precise scale whose causal origin is being matched. These are bounds in a simplified model, not a measurement of how many e-folds nature produced.

(e) A shrinking comoving Hubble radius lets a region that was initially causally connected expand to encompass the scales that later re-enter our horizon. This addresses the horizon problem. Quantum fluctuations are stretched as well, so inflation generically leaves small perturbations rather than perfect smoothness; those perturbations are essential seeds for later structure.

Checks. More e-folds make the universe flatter, not more curved. The later growth requirement increases the required N. The factor is e^{-2N}, not e^{-N}, because curvature appears divided by $(aH)^2$.

Why this is hard. The calculation is short once the correct variable is identified. The challenge is to distinguish the curvature constant, the curvature density parameter and the observable fractional deviation, and to keep the post-inflationary history in the budget.

Final review: the ideas worth retaining

1. **Expansion is a changing relation between coordinates and proper distances.** It is not an explosion from a central point.
2. **Redshift is a ratio of scale factors.** Velocity, lookback time and distance require additional definitions.
3. **Gravity responds to pressure as well as energy density.** This is why radiation and Λ behave so differently.
4. **The CMB is a thermal record, not merely a faint glow.** Its spectrum, anisotropies and relation to light elements test different parts of the same history.
5. **Horizons are integrals over causal paths.** The Hubble radius is not a universal boundary.
6. **A difficult solution should end with checks.** Limits, units, signs and alternative derivations are how you know that the algebra describes the intended universe.

Return to the Lecture + Problems, repeat a challenge without looking at the derivation, and try to explain the result to another student in ordinary language.